UNIT 2 · TOPIC 12(B)

Drone Regulations and Government Schemes

Syllabus topic: Selection and operation of drones

Drone Rules 2021 — weight categories (all-up weight)

Class	Weight
Nano	<250 g
Micro	250 g–2 kg
Small	2–25 kg (agricultural spray drones)
Medium	25–150 kg
Large	>150 kg

Digital Sky platform (DGCA)

Handles type certification, unique identification number (UIN) registration, and the airspace map:

Zone	Rule
Green	Fly up to 120 m (400 ft) without permission
Yellow	ATC permission needed
Red	No-fly

- A **Remote Pilot Certificate (RPC)** from a DGCA-authorised Remote Pilot Training Organisation is required to operate small and larger drones (and commercial micro drones). It is **not** needed for nano and non-commercial micro drones.
- Import of built-up drones was banned (**Feb 2022**) to promote Make-in-India (components/R&D exempt); the **PLI scheme** supports drone manufacturing.

Government promotion (verified July 2026)

SMAM "Kisan Drone" beneficiary	Assistance
ICAR institutes, KVKs, SAUs, FMTTIs (demonstrations)	100% of drone cost, up to ₹10 lakh
FPOs (demonstrations)	75% grant
Custom Hiring Centres — cooperatives, FPOs, rural entrepreneurs	40%, up to ₹4 lakh
Custom Hiring Centres — agriculture graduates	50%, up to ₹5 lakh
Individuals — SC/ST, small & marginal, women and NE farmers	50%, up to ₹5 lakh
Individuals — others	40%, up to ₹4 lakh

Namo Drone Didi (Central Sector Scheme, ₹1,261 crore, 2023-24 to 2025-26): drones to **15,000 women Self-Help Groups** at **80% subsidy up to ₹8 lakh**, with the balance via an AIF loan at 3% interest; 15-day training (one pilot + one technician per SHG); target ≥ ₹1 lakh/yr additional SHG income from custom rental spraying.

Scheme slabs change — check agriwelfare.gov.in before quoting them.

Practice MCQs

Q204. Under Drone Rules 2021, agricultural spray drones (typically 2–25 kg) fall in the class:

(a) Nano
(b) Micro
(c) Small
(d) Medium

Q205. A NANO drone weighs:

(a) <250 g
(b) <2 kg
(c) <25 kg
(d) <150 kg

Q206. India's online platform for drone registration, UIN and airspace maps is:

(a) UMANG
(b) Digital Sky (DGCA)
(c) DigiLocker
(d) AGMARKNET

Q207. In a GREEN zone, drones may operate without permission up to:

(a) 60 m
(b) 120 m (400 ft)
(c) 500 m
(d) Unlimited

Q208. A Remote Pilot Certificate is NOT required for:

(a) Small drones
(b) Medium drones
(c) Nano and non-commercial micro drones
(d) Any drone

Q213. Under SMAM, ICAR institutes/KVKs/SAUs get drone assistance of:

(a) 40%
(b) 50%
(c) 75%
(d) 100% up to ₹10 lakh

Q214. Namo Drone Didi provides drones to women SHGs at a subsidy of:

(a) 40% up to ₹4 lakh
(b) 50% up to ₹5 lakh
(c) 80% up to ₹8 lakh
(d) 100%

Answers

Q204 — (c) Small — *Drone Rules 2021 — nano <250 g, micro ≤2 kg, small ≤25 kg, medium ≤150 kg, large >150 kg.*

Q205 — (a) <250 g

Q206 — (b) Digital Sky (DGCA)

Q207 — (b) 120 m (400 ft) — *Green zone ceiling 120 m.*

Q208 — (c) Nano and non-commercial micro drones — *RPC exemption only for nano and non-commercial micro drones.*

Q213 — (d) 100% up to ₹10 lakh — *SMAM slabs — 100%/₹10 lakh (ICAR/KVK/SAU), 75% FPO demos, CHC 40%/₹4 lakh (50%/₹5 lakh agri graduates), individuals 40–50%.*

Q214 — (c) 80% up to ₹8 lakh — *Namo Drone Didi — 80%/₹8 lakh, 15,000 women SHGs, ₹1,261 crore (2023-24 to 2025-26).*

Sources: Kepner et al. (hydraulic principles); Drone Rules 2021 (Ministry of Civil Aviation); MoA&FW SOPs for drone application of pesticides & nutrients; SMAM drone guidelines & PIB releases (Kisan Drone, Namo Drone Didi — verified July 2026); ISO 11783 (ISOBUS); NPTEL lectures on mechatronics & precision agriculture.